Aayush Panda

aayush.vinayak@gmail.com | linkedin.com/in/aayush-panda | github.com/AayushPanda

Technical Skills

Programming Languages: C, C++, Rust, Python, Java, TypeScript, SQL, Haskell, Scheme, Bash, HTML/CSS, x86 assembly
Machine Learning: Transformers, RNNs, flow-matching, VAEs, PyTorch, Tensorflow, Sklearn, Numpy
Databases: SQL dbs, Firestore, Neo4J, Chroma

Education

University of Waterloo — **GPA:** 3.8/4.0
Bachelor of Computer Science, Honours Co-op — *Sep. 2024 – Apr. 2028*

Experience

ML & NLP Engineering Intern — Sep. 2025 – Apr. 2026
Huawei Canada

- Developed agents and encoder-only models for voice/drawn gesture productivity software use (**10 million users)**
- Improved performance of Neo4j knowledge graph updates using Merkle trees. Reduced latency by **95%**
- Developed text indexing algorithms for AI text editor with consistency guarantees ($\mathbf{F_1 = 0.95}$ chunk retrieval for $100+$ pages)
- Adapted a hybrid ASR model architecture to work with natural language context to guide transcription, reduced WER by $\sim \mathbf{5\%}$
- Developed iteratively self-improving coding agents, with outputs being preferred to production code in $\mathbf{90\%}$ of out-of-dist. tasks

Undergraduate Researcher — Apr. 2025 – Aug. 2025
University of Waterloo Multi-Sensory Brain and Cognition Lab (MBC)

- Research on how batsmen's eyes track baseballs, measuring visual perception bandwidth, and correlating with batting ability
- Developed hybrid/synthetic data pipelines to train a model to track baseballs in captured video with $\mathbf{97\%}$ success rate
- Developed correspondence algorithm to synchronise data from high speed head mounted camera and eye tracking data

Undergraduate Researcher — Apr. 2025 – Aug. 2025
University of Waterloo Data Privacy Lab

- Research on private record linkage protocols using locality sensitive hashing, supervised by Dr. Florian Kerschbaum
- Worked on alternatives to frequency-smoothing as in Wei and Kerschbaum 2023 offering $\mathbf{5-10x}$ speedup on testcases
- Developed novel private set intersection algorithm using k-d cuckoo hashtables, reducing runtime from $\mathbf{O(n^2)}$ to $\mathbf{O(n)}$

Software Engineering Intern — Mar. 2025 – Apr. 2025
Toma (A16Z, YC W25)

- Architected an AI lead generation engine that helped scale Toma from **20 to 250+** customers, driving ARR from **$500K to $3M**
- Reverse-engineered dealership booking platforms to build the first-ever dataset of all **25,000** North American car dealerships
- Built a call orchestration system deploying **300,000** voice AI agents to collect data on all NA dealerships' phone service quality

Projects

Concord — *Rust, Vulkan, CUDA, HLSL*

- Cross-platform runtime in Rust for accelerated computation, with a unified ISA and memory model for CPU/GPU operations.
- Architected bytecode ISA and VM runtime with native cooperative multitasking and FFI
- Developed modular compilation toolchain with inline bytecode support to allow for elegant native extensions
- Created CLI interactive debugger with VM state inspection and variable watching

Semantify — *Python, React, FastAPI, Sentence Transformers, UMAP, Ollama*

- Built a system that semantically organises files into a directory structure ($1000+$files$/s$) using embedding models and custom hierarchical clustering and subtree merging algorithms
- Developed an interactive visualization of document embeddings to allow intuitive semantic exploration
- Implemented a RAG-powered chat interface to intelligently query uploaded documents with source citation

Phased Array SONAR — *C++, Python, Xtensa LX6 microprocessor, AVR RISC processors*

- Engineered a sub-degree precision beam-steering phased SONAR array with real-time radar-style display on oscilloscopes.
- Designed an acoustic waveguide to suppress grating lobes, enhancing beam directivity.
- Built a phased array simulator suite to visualize beamforming, steering, and beam focussing behaviors in 2D.

Patents

CA 3222437: Device for redirection of optical beams using virtual gratings generated by stationary waves.
CA 3119717: Compliant mechanism for operating flight control surfaces of a remotely piloted aircraft.

Awards

University of Waterloo President's Research Award
Jane Street Estimathon @ UWaterloo (2024): First place
Hack the North 2022: Winner (out of 829 participants)
PicoCTF: 2nd place in Canada, 14th (top 0.001%) globally